

$$G \times A = M + \check{C} - A$$

Robot Overview

Our robot uses a dual-processor architecture combining a Raspberry Pi 5 and a Teensy 4.1 for high-level strategy and real-time hardware control.

The robot uses:

- omnidirectional drive with four self-made omniwheels
- sensor fusion for ball localization
- computer vision for goal detection
- Raspberry Pi as a powerful debugging tool

Mechanical Design

- Fully custom 3D-printed chassis
- Designed in **OpenSCAD** and FreeCAD
- Modular internal layout for rapid repairs and upgrades
- Omnidirectional drivetrain with four independently controlled wheels

Software

Vision & Navigation

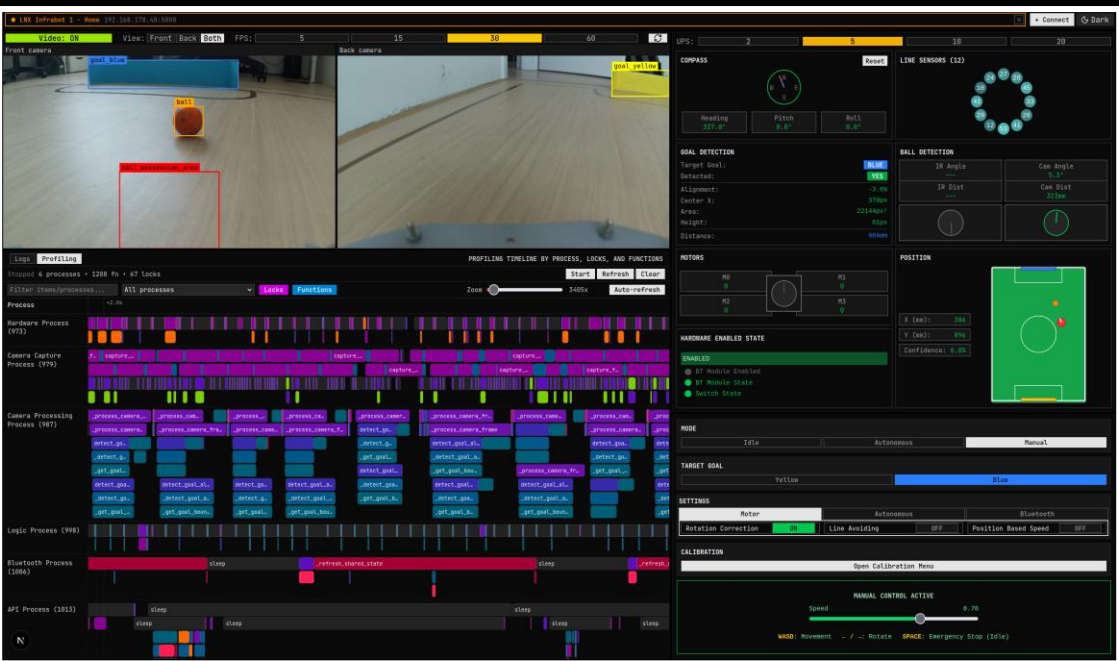
The Raspberry Pi 5 performs real-time computer vision and autonomous navigation during gameplay. Using OpenCV-based image processing together with infrared sensing, the robot continuously detects:

- the ball,
- field goals,
- and its orientation on the field.

Sensor fusion combines:

- camera localization,
- IR ball tracking,
- and BN0055 IMU orientation data

to maintain stable movement and accurate field awareness during matches. The omnidirectional drivetrain allows the robot to move in any direction while continuously maintaining orientation toward the target goal and keeping control of the ball. Real-time decision logic calculates movement, rotation, and positioning at up to 120 updates per second.



The custom dashboard provides real-time visualization of camera feeds, sensor data, position detection, a profiling system and a calibration system. Using WebSocket communication with the Raspberry Pi, the interface enables low-latency monitoring, debugging, and remote control of the robot.

LNx ROBOTS

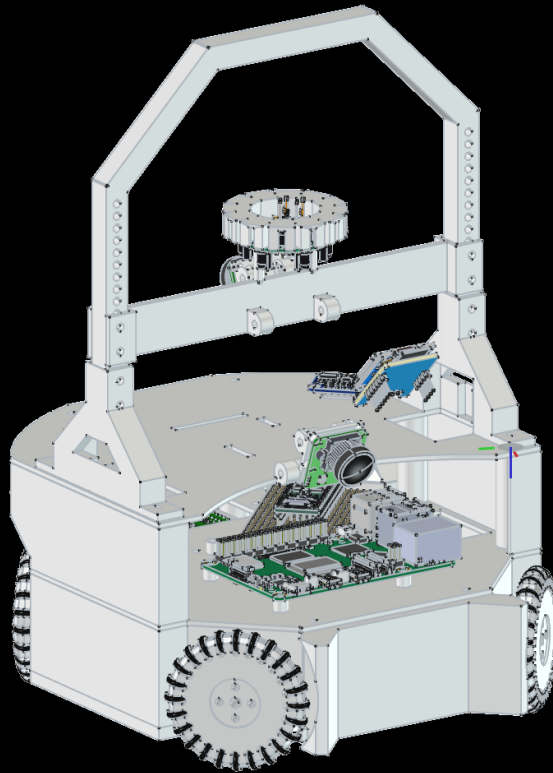
INFRARED

About us

Gymnázium, Grösslingová 18, Bratislava, Slovakia.

LNX InfraBot

- Fully custom autonomous soccer robot designed for RoboCupJunior Soccer Infrared. Designed and developed since November 2025



Hardware

Main Components

- Raspberry Pi 5
- Teensy 4.1
- 4× Pololu 370 Metal Gearmotors
- 2× Pololu Dual G2 Motor Drivers
- BN0055 IMU
- 2× Arducam cameras
- MRM IR Finder 3
- OLED status display

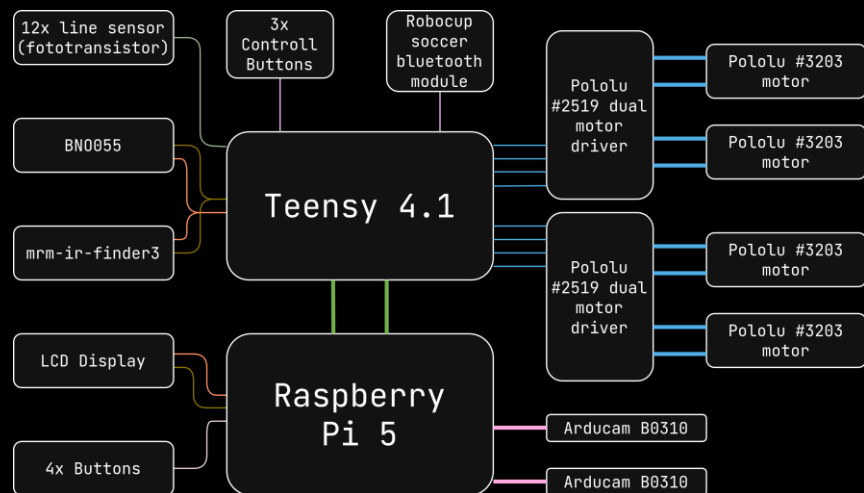
Real-Time Control System

The Teensy 4.1 handles:

- motor control
- sensor communication
- line sensor readings (~4700x/s)
- low-latency hardware processing

Separating hardware control from high-level processing improves responsiveness and system stability during gameplay.

Hardware Architecture



Slovnaft



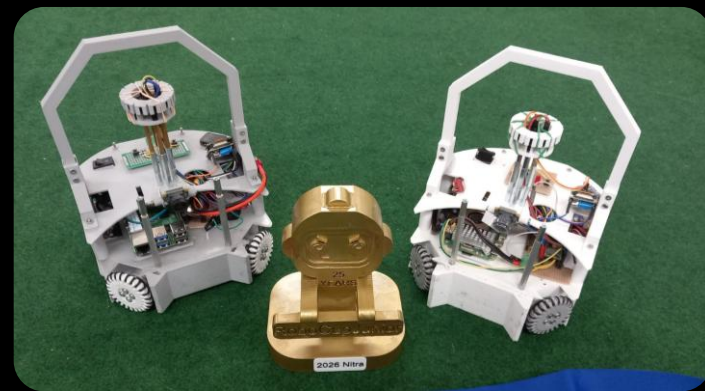
MAKING CLOUD EASY



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NÁRODNÝ INŠTITÚT VZDELÁVANIA A MLÁDEŽ



GitHub

Exploded View

